

Problem Set - 3

1) $\therefore \frac{dx}{dt} = -500x + xy$

$\frac{dy}{dt} = -600y + 3xy$

$\therefore x(-500 + y) = 0$

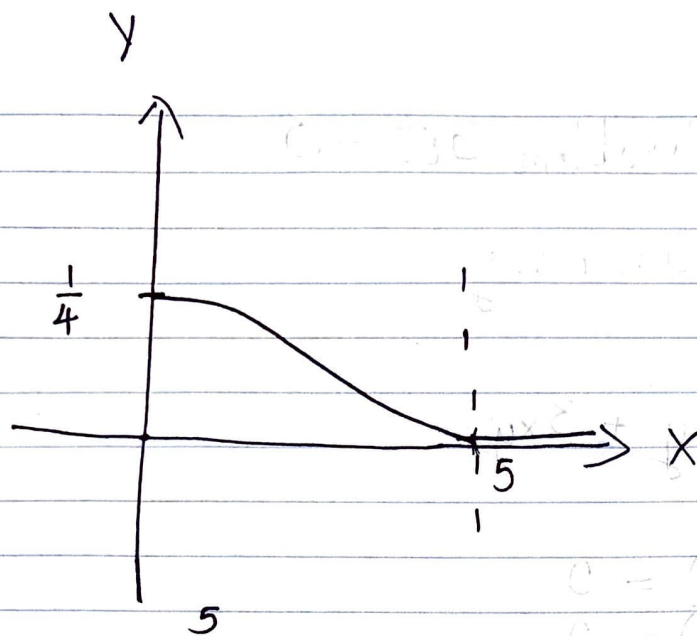
$y(-600 + 3x) = 0$

$\therefore y = 500 - i)$

$x = 200 - ii)$

2) $\therefore \frac{dy}{dx} = x^2 + y^4$

Setting it equal to 0, the derivative must change from positive to negative. But since x^2 and y^4 are always positive, the derivative is not going to be negative. Hence, the model has the property we are looking for.



$$\therefore V = 2\pi \int_0^5 x \cdot \left(\frac{1}{x^2 + 4} \right) \cdot dx$$

$$\text{let } x^2 + 4 = u$$

$$2x = \frac{du}{dx}$$

$$= \pi \int_0^{29} \frac{du}{u}$$

$$= \pi \cdot [\ln u]_0^{29}$$

$$= \pi [\ln(x^2 + 4)]_0^5$$

$$= \pi (\ln(29) - \ln(4))$$

$$= \pi \ln(29) - \pi \ln(4) = \pi \ln\left(\frac{29}{4}\right)$$

$$b) \therefore y = \frac{1}{x^2 + 4}$$

$$x^2 + 4 = \frac{1}{y}$$

$$\leftarrow \text{definizione di lunghezza arco di una funzione}$$

$$x^2 = \frac{1}{y} - 4$$

$$x = \sqrt{\frac{1}{y} - 4}$$

$$\therefore V = \pi \int_0^{1/29} (5)^2 \cdot dy + \pi \int_{1/29}^{1/4} \left(\sqrt{\frac{1}{y} - 4} \right)^2 \cdot dy$$

$$= \pi \left[\left[(25y) \right]_0^{1/29} + \left[\ln \left| \frac{1}{y} - 4 \right| \right]_{1/29}^{1/4} \right]$$

$$= \pi \left[\left(\frac{25}{29} - 0 \right) + \left[\left(\ln \left| \frac{1}{4} \right| - \ln \left| \frac{1}{29} - 4 \right| \right) \right] \right]$$

$$= \pi \left(\frac{25}{29} - 1 + \ln \left| \frac{1/4}{1/29} \right| \right)$$

$$= \pi \cdot \ln \left| \frac{29}{4} \right|$$

$$\left(\frac{0}{1} + \left(\frac{e}{e} \right)^{\ln \left(\frac{1}{4} \right)} \right) = \left(\frac{x}{4} + \left(\frac{x}{e} \right)^{\ln \left(\frac{1}{4} \right)} \right) \cdot \frac{1}{x}$$

4.

$$a) V = \pi \int_0^5 \left(\frac{1}{x^2 + 4} \right)^2 \cdot dx$$

Creating the above integral as indefinite $\rightarrow$

$$\pi \int \frac{1}{(x^2 + 4)^2} \cdot dx$$

$$\text{let } x = 2 \tan \theta$$

$$dx = 2 \sec^2 \theta \cdot d\theta$$

$$= \pi \int \frac{1}{(4 \sec^2 \theta)^2} \cdot 2 \sec^2 \theta \cdot d\theta$$

$$= \pi \int \frac{1}{8 \sec^2 \theta} \cdot d\theta$$

$$= \frac{\pi}{8} \int \frac{(1 + \cos 2\theta)}{2} \cdot d\theta$$

$$= \frac{\pi}{16} (\theta + \sin \theta \cdot \cos \theta) + C$$

$$= \frac{\pi}{16} \left(\tan^{-1} \left(\frac{x}{2} \right) + \frac{2x}{x^2 + 4} \right) \Bigg|_0^5 = \frac{\pi}{16} \left(\tan^{-1} \left(\frac{5}{2} \right) + \frac{10}{29} \right)$$

$$b) \therefore y = \frac{1}{x^2 + 4}$$

$$x = \sqrt{\frac{1}{y} - 4}$$

$$\text{at } x = 0, y = \frac{1}{4}$$

$$\text{at } x = 5, y = \frac{1}{29}$$

$$V = \int_{\frac{1}{29}}^{\frac{1}{4}} y \, dy$$

$$\therefore V = 2\pi \left[\int_0^{1/29} 5y \, dy + \int_{1/29}^{1/4} y \cdot \sqrt{\frac{1}{y} - 4} \, dy \right]$$

$$5. \therefore y = \sin\left(\frac{\pi x}{9}\right)$$

$$\frac{dy}{dx} \text{ or } y' = \frac{\pi}{9} \cdot \cos\left(\frac{\pi x}{9}\right)$$

$$\therefore \int_0^{22.5} \sqrt{1 + \left(\frac{\pi}{9} \cdot \cos\left(\frac{\pi x}{9}\right)\right)^2} \, dx$$

$$\Delta x = \frac{22.5 - 0}{5} = 4.5$$

$$f(x_i^*) = \sqrt{1 + \frac{\pi^2}{2 \times 81}}$$

$$= \sqrt{1 + \frac{\pi^2}{162}}$$

$$L = 22.5 \sqrt{1 + \frac{\pi^2}{162}}$$

$$= 23.1753$$

6.

$$a) \therefore y = C + A \cosh\left(\frac{\pi x}{A}\right)$$

$$y' = 0 + \sinh\left(\frac{x}{A}\right)$$

$$\therefore L = \int_{-8}^8 \sqrt{1 + \left(\sinh\left(\frac{x}{A}\right)\right)^2} \cdot dx$$

$$= \int_{-8}^8 \sqrt{\cosh^2\left(\frac{x}{A}\right)} \cdot dx$$

$$= \int_{-8}^8 \cosh\left(\frac{x}{A}\right) \cdot dx$$

$$= \left[A \sinh\left(\frac{x}{A}\right) \right]_{-8}^8$$

$$= A \sinh\left(\frac{8}{A}\right) - \left(\sinh\left(-\frac{8}{A}\right) \right)$$

$$\left[\because \sinh(-x) = -\sinh(x) \right]$$

$$= 2A \sinh\left(\frac{8}{A}\right)$$

b) $\therefore 2B = 80$
 $B = 40$

$$\therefore L = 81$$

$$\therefore 81 = 2A \sinh\left(\frac{40}{A}\right), \quad A \approx 146.3325$$

The lowest point occurs at $x=0 \rightarrow A =$

$$\therefore y(0) = C + A \cosh(0) = 25$$

$$\therefore C + A = 25$$

$$C = 25 - A$$

$$y(40) = C + A \cosh\left(\frac{40}{A}\right)$$

$$= 25 - A + A \cosh\left(\frac{40}{A}\right)$$

$$25 - A \left(1 - \cosh\left(\frac{40}{A}\right)\right)$$

$$= 30.50 \text{ ft tall}$$